Supplementary Material: E-Waste Safety: Hierarchical CNN with Adaptive Fuzzy Temporal Supervisory Risk Model for Pre-Shred Ignition Control

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Note—This supplement holds sections, tables and a figure moved without change from the full-length version of the paper to meet the page limit. Numbers with the prefix S refer to this supplement; numbers without it refer to the main paper. References are numbered separately.

S1. CFNC: Catastrophic False Negative Cost

A plain miss rate treats every missed time step equally, even though missing a cell during runaway (Phase 3) is more serious than missing it during early self-heating (Phase 1). CFNC, or Catastrophic False Negative Cost, gives misses different weights according to their phase:

$$CFNC = (10 \cdot N_run + 2 \cdot N_miss) / N_haz \quad (4)$$

N_haz counts all time steps in Phases 1–4. N_miss counts hazardous steps when the system remained in Normal, while N_run counts missed steps during runaway and fire (Phases 3 and 4). A system that misses only in the early phases receives a CFNC equal to twice its Hazard FNR; each runaway-phase miss adds another 10/N_haz. We selected the weights 10 and 2 by hand rather than fitting them to data. In the single-item results below, runaway-phase misses are zero or negligible (no more than 0.0001 for B1). CFNC is therefore almost exactly twice Hazard FNR in those cases and provides no separate evidence. It distinguishes systems only when runaway-phase misses occur, as in the blinded vision proxy test (Section S6).

CFNC assigns weight 10 to runaway-phase misses and 2 to early-phase misses. We recalculated it with weights (5, 1), (20, 1), (2, 1), and (1, 1). The system ranking did not change because none of the systems missed a runaway-phase step in the single-battery scenario.

S2. CNN Classification Results

Figure S1 shows the confusion matrices for the coarse and fine classifiers.

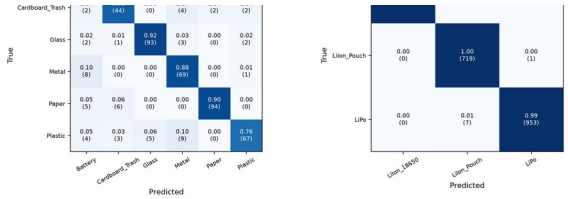


Fig. S1. Confusion matrices for the coarse six-class model using the battery threshold (left, 692 test images) and the fine three-class subtype model (right, 2,208 test images)

S3. Baseline Comparison

Table S1 compares the four systems in a single run.

TABLE S1. Baseline results for one run (seed 42): Hazard FNR, seconds from battery arrival to Warning and Emergency Stop, and CFNC

System	Hazard FNR	T_warn (s)	T_stop (s)	CFNC
B1: Vision proxy	0.0688	144.2	161.6	0.1375
B2: Rule-based	0.0008	91.7	203.3	0.0016
B3: Fixed Mamdani FIS	0.0675	150.8	193.9	0.1351
Proposed	0.0000	0.0	191.1	0.0000

S4. Statistical Robustness Across Sensor-Noise Realisations

Table S1 uses one deterministic draw of the hazardous scenario (seed 42). To check whether its results depend on that draw, we ran the same scenario with 30 independent sensor-noise realisations (seeds 100–129). We regenerated temperature, gas, and metal-detection readings from the same underlying physics model each time and kept the scenario timeline fixed, including battery arrival at t = 20 s. Table S2 gives the mean and standard deviation across these runs. For T_warn and T_stop, the table excludes the few runs in which a system never crossed the relevant threshold during the 400 s simulation: 2–3 of 30 runs per system.

TABLE S2. Mean and standard deviation over 30 sensor-noise realisations. T_warn and T_stop include only runs that crossed the threshold: B1 28/28, B2 28/27, B3 28/27, Proposed 30/27

System	Hazard FNR	T_warn (s)	T_stop (s)	CFNC
B1: Vision proxy	0.0993 ±0.0569	180.4 ±63.8	196.6 ±64.5	0.1986 ±0.1137
B2: Rule-based	0.0015 ±0.0013	102.9 ±58.6	239.5 ±58.6	0.0029 ±0.0026
B3: Fixed Mamdani FIS	0.1194 ±0.0798	188.7 ±63.3	230.9 ±58.0	0.2388 ±0.1596
Proposed	0.0060 ±0.0153	27.1 ±64.8	228.1 ±58.3	0.0119 ±0.0306

All four systems used the same 30 sensor-noise draws, so paired tests compare their behaviour under identical conditions. The Proposed system had a lower Hazard FNR than B3 (0.0060 vs 0.1194; mean reduction 0.1134; paired $t(29) = -7.69$, $p < 0.0001$). It was lower in 28 of 30 seeds and higher in none. It also had a lower rate than B1 (mean reduction 0.0933; $t(29) = -8.71$, $p < 0.0001$). CFNC showed the same pattern, but in these runs it is essentially twice Hazard FNR (Section S1), so it is not independent evidence. The result persists after excluding the three seeds in which the simulated cell never reached runaway (107, 122, and 129): with $n = 27$, the reduction was 0.1095 ($p < 0.0001$). B2 had the lowest raw Hazard FNR (0.0015), and its difference from the Proposed system was not significant ($t(29) = 1.63$, $p = 0.11$; the Proposed system was lower in 21 seeds and higher in 5). We therefore do not claim that the Proposed system reduces misses relative to B2 in this scenario. B2's low miss rate comes from applying fixed thresholds directly to raw sensors, which also causes it to stop during battery-free gas bursts (Section V-C). The camera allowed the Proposed system to issue a Warning on arrival in 24 of 30 seeds; median T_warn was 0 s.

S5. CNN-Fusion Ablation and Alpha Sensitivity

To measure the effect of the CNN fusion term in Eq. (3), we repeated the 30-seed sweep with $\alpha = 0$. At $\alpha = 0.35$, Hazard FNR decreased from 0.0412 to 0.0060 (mean reduction 0.0352; paired $t(29) = -6.43$, $p < 0.0001$), with a lower value in 23 of 30 seeds. The main change was that recognising the battery raised fused risk above the Warning threshold on arrival. Mean T_warn was 27 s at $\alpha = 0.35$, compared with 176 s when $\alpha = 0$.

Table S3 shows the Hazard FNR as α increases from 0 to 1.5. It reaches a plateau at $\alpha = 0.2$ and does not improve beyond that point. Higher α values bring Emergency Stop forward: the mean time is 228.1 s at $\alpha = 0.35$ and 190.1 s at $\alpha = 1.5$. We therefore use $\alpha = 0.35$, which is on the plateau.

TABLE S3. Effect of α across the same 30 hazardous-scenario seeds as Table S2 (mean ± standard deviation)

α	Hazard FNR	T_stop (s)	CFNC
0.0	0.0412 ±0.0264	229.1 ±58.3	0.0824 ±0.0529
0.1	0.0291 ±0.0233	229.1 ±58.3	0.0583 ±0.0466
0.2	0.0060 ±0.0153	228.4 ±58.3	0.0119 ±0.0306
0.35 (paper)	0.0060 ±0.0153	228.1 ±58.3	0.0119 ±0.0306
0.5	0.0060 ±0.0153	228.1 ±58.3	0.0119 ±0.0306
0.75	0.0060 ±0.0153	199.4 ±64.8	0.0119 ±0.0306
1.0	0.0060 ±0.0153	196.4 ±64.3	0.0119 ±0.0306
1.5	0.0060 ±0.0153	190.1 ±65.0	0.0119 ±0.0306

We also tested whether a high α could let an uncertain CNN prediction for a benign item cause a false stop. We used two battery-free scenarios, each with 30 seeds: ambient conditions only, and eight short gas spikes per run. No false Emergency Stop occurred for any tested α up to 1.5. In the gas-spike scenario, peak fused risk remained at 0.50, below the Critical threshold of 0.55. The retrained coarse model usually assigns benign items a low battery probability, leaving little for fusion to amplify.

The tested scenarios show no false-alarm cost from increasing α. We retain 0.35 because it lies on the detection plateau and gives the same stop time as values from 0.2 to 0.5. Future work should tune α across a broader range of failure scenarios.

S6. Multi-Item Conveyor and Camera Occlusion

Sections S3 to S5 examine one battery on an otherwise empty belt. We had discussed two other conditions in Section VI, a second object ahead of the battery and a blinded camera, without testing either one. We therefore ran both tests across the same 30 seeds (100–129).

For the multi-item test, a benign metal object crossed the belt from t = 15 to 45 s. The battery arrived at t = 60 s, after a gap longer than the 3 s reset time. Across all 30 seeds, none of the four systems raised a Warning, Critical, or Emergency_Stop during the benign object's passage (Table S4, rightmost column). The Proposed system then missed no hazardous steps (Hazard FNR = 0.0000). The corresponding rates were 0.1049 for B3 ($p < 0.0001$; the Proposed system was lower in 28 of 30 seeds), 0.0890 for B1 ($p < 0.0001$), and 0.0010 for B2 (paired $t(29) = -6.14$, $p < 0.0001$; the Proposed system was lower in 24 of 30 seeds). Its mean stop time was 227.8 s after the battery arrived. We did not test a third item after the battery because the Emergency Stop is designed to remain latched until manual reset.

TABLE S4. Multi-item test with a benign metal object followed by a battery: mean and standard deviation over 30 seeds

System	Hazard FNR	T_stop (s)	Alerts on benign item
B1: Vision proxy	0.0890 ±0.0476	197.8 ±65.8	0 / 30 seeds
B2: Rule-based	0.0010 ±0.0009	239.5 ±58.6	0 / 30 seeds
B3: Fixed Mamdani FIS	0.1049 ±0.0596	231.3 ±57.8	0 / 30 seeds
Proposed	0.0000 ±0.0000	227.8 ±58.3	0 / 30 seeds

To simulate camera occlusion, we set P_battery = 0 throughout the run. This is equivalent to $\alpha = 0$ because the temporal state does not otherwise use P_battery. The Proposed system then relies on sensor risk alone. Hazard FNR increases from 0.0060 to 0.0412 (paired $t(29) = 6.43$, $p < 0.0001$), reflecting the loss of camera information measured in Section S5. It remains below B3's rate of 0.1194 (reduction 0.0782; $t(29) = -5.80$, $p < 0.0001$; lower in 26 of 30 seeds). B2 has a lower rate (0.0015, $p < 0.0001$) because its fixed thresholds use raw sensor readings and do not depend on the camera. B1 is blind by definition when the camera is occluded. Its Hazard FNR is 1.0 across the 28 runs where the cell reached a hazardous phase, or 0.9333 if the two runs that never reached a hazardous phase are counted as zero misses. We include that row for completeness.

TABLE S5. Camera-occlusion test (P_battery = 0): mean and standard deviation over 30 seeds

System	Hazard FNR	T_stop (s)	Reached E-stop
Proposed (camera working)	0.0060 ±0.0153	228.1 ±58.3	27 / 30
Proposed (occluded)	0.0412 ±0.0264	229.1 ±58.3	27 / 30
B3: Fixed Mamdani FIS	0.1194 ±0.0798	230.9 ±58.0	27 / 30
B2: Rule-based	0.0015 ±0.0013	239.5 ±58.6	27 / 30
B1: Vision proxy (occluded)	1.0000 (n = 28)	never	0 / 30

When debris covers a battery, P_battery is approximately zero. Section S6 tests this condition across 30 seeds. The Proposed system still detects risk from the sensors and has a lower Hazard FNR than B3 (0.0412 vs 0.1194, $p < 0.0001$), although it loses the benefit of CNN fusion shown in Section S5. A vision-only system has no input in this test. The simulation Hazard FNR is different from the coarse CNN test-set FNR (0.0000): one measures missed hazardous time under dynamic conveyor conditions, while the other measures image classification on a held-out set. B2 again has the lowest raw miss rate because it thresholds the sensor readings directly. The health monitor also flags a lost camera image as a fault (Section S10).

In the 30-seed multi-item test (Section S6), a benign metal object passed ahead of the battery. It caused no false alarm and did not delay Emergency Stop for the battery: the mean was 227.8 s, close to 228.1 s in the single-item test. We did not simulate a third item after the battery because the latched Emergency Stop requires manual reset.

S7. Thermal Model Compared with Published Data

The results so far rely on a simulated temperature profile. We therefore compared its main temperature landmarks with published measurements (Table S6). This is a landmark comparison only; we did not fit the simulation to a dataset. We did not compare gas concentrations because the published data report gas amount per cell (up to 0.27 mol for 18650 cells [1]), which cannot be converted to sensor ppm without the enclosure volume. Section S11 examines gas composition and timing separately.

TABLE S6. Temperature landmarks in the simulation and published studies

Landmark	Published	Model phase threshold	Agreement
SEI decomposition onset	about 90°C (80–120°C) [2]	45°C (Phase 1 start)	Earlier in simulation
Separator melting	about 130°C [2]	H2 onset 70°C, Phase 2 end 90°C	Earlier in simulation
Cathode decomposition	above about 180°C [2]	Rapid rise (Phase 3) from 90°C	Earlier in simulation
Peak cell temperature	up to about 850°C [1]	851.2°C	Consistent

The simulated peak temperature is close to the published value, but its onset temperatures are about 40–90°C lower than reported ranges, which vary with cell chemistry and heating method. As a result, the simulated cell reaches alarm thresholds earlier than a real cell may. The T_warn and T_stop values in Tables S1–S5 describe this simulation; they are not predictions for a real conveyor. We chose the lower thresholds as a conservative setting, but did not validate them. Since every system receives the same simulated data, the comparisons remain controlled. Sections S10 and S11 repeat the main tests using a model with published onset temperatures. Fitting the phenomenological model to measured cell data remains future work.

S8. Design of the Adaptive Memory, Noise Tests and Already-Failing Cells

With CNN fusion enabled, the hard-noise results show the practical difference from B2 (Table S7). B2 makes a false Emergency Stop in all 30 runs, and B3 produces a false Critical state in all 30. The Proposed system does neither. In the harder-noise test, it reaches Critical in 14 runs and Emergency Stop in 1 of 30; B2 stops in all 30.

TABLE S7. Full-system results on held-out seeds: false escalations in battery-free noise and mean time to Emergency Stop for cells already failing

System	False Critical / E-stop (hard)	False E-stop (harder)	Stop (s) 60°C ×3 / 80°C ×6 / 90°C ×10
B1: Vision proxy	0 / 0	0	8.7 / 1.8 / 0.0
B2: Rule-based	0 / 30	30	13.9 / 1.8 / 0.0
B3: Fixed Mamdani FIS	30 / 0	30	11.2 / 1.5 / 5.1
Proposed	0 / 0	1	10.2 / 3.1 / 3.1

S9. Dwell Times, Seed Repeats, Backbones and Cluttered Frames

The controller waits 2 s in Warning and 1 s in Critical before escalating. This explains why B2 stops faster for rapidly failing cells (Table S7). Table S8 shows what happens when we shorten the dwell times. For a cell arriving at 80°C, halving the dwell reduces the stop time from 3.1 s to 1.6 s; removing it reduces the time to 0.2 s. False escalations during hard noise rise from 0 to 2 and 11 of 30 runs, respectively. A 0.5 s / 0.2 s setting produces false escalation in 9 runs. We retain 2 s / 1 s, although 14 of 30 harder-noise runs still reach Critical and one reaches Emergency Stop. The harder bursts exceed the rule base's 800 ppm H2 hazard threshold; the retuned gate in Section V-D removes these escalations. In a plant, dwell time should reflect how long a failing cell takes to reach the shredder. A 3 s stop is useful only if the cell is still more than 3 s from the inlet. In our line model (Section V-E), the hood ends 1.0 m from the camera and the inlet is 2.5 m away. At 0.25 m/s, an item leaving the hood is 6 s from the inlet. Warning reduces belt speed to 60%, and every already-failing cell was diverted or stopped before the inlet (Table VII).

TABLE S8. Dwell-time sensitivity for Proposed on held-out seeds: mean time to Emergency Stop for already-failing cells and false escalations in 30 hard-noise runs

Dwell (Warning / Critical)	Stop 60 / 80 / 90°C (s)	False esc. hard	False esc. harder
2 s / 1 s (paper)	10.2 / 3.1 / 3.1	0	14
1 s / 0.5 s	10.2 / 1.6 / 1.6	2	16
0.5 s / 0.2 s	10.2 / 0.7 / 0.7	9	30
none	10.2 / 0.2 / 0.2	11	30

EfficientNet [3] scales network depth, width, and resolution together. Its smallest model, B0, offers ImageNet

accuracy with relatively low inference cost. MobileNetV2 [4], which uses inverted residual blocks, is a lighter option. Section S9 compares both models with EfficientNet-B3 on leak-free data. We retain B0 because it detected every intact test battery in every run and its measured latency (Section IV-A) fits the 100 ms control budget.

We retrained the coarse classifier with additional seeds and two alternative backbones, MobileNetV2 [4] and EfficientNet-B3 [3], using the same data splits (Section IV-A, Table S9). Each model received its own battery threshold, selected on its validation set using the production model's rule. EfficientNet-B0 with seed 42 reproduces the reported result: 91.18% accuracy, threshold 0.1113, and battery recall of 99.25%. Across three EfficientNet-B0 runs, mean accuracy is 90.80% ± 0.55 percentage points, and no run misses an intact test battery.

In one run, EfficientNet-B3 reaches 93.06% accuracy. It misses at most one intact test battery, produces an average of 16.0 false alarms, has 10.7 M parameters, and takes 20.6 ms per image on the GPU and 43.5 ms on the CPU. MobileNetV2 reaches 91.52% ± 0.58 across three runs, misses at most two intact test batteries, and produces an average of 13.7 false alarms. It has 2.2 M parameters and takes 8.3 ms per image on the GPU and 16.9 ms on the CPU. These latency measurements were taken while other jobs shared the machine. EfficientNet-B3's single run is 2.26 percentage points above the EfficientNet-B0 mean and has fewer false alarms, but we did not test that one-run difference for significance. EfficientNet-B0 is the only backbone to detect every intact test battery in every run; EfficientNet-B3 misses up to one, and MobileNetV2 up to two. On the seven damaged test batteries, EfficientNet-B0 detects four or five, EfficientNet-B3 detects six, and MobileNetV2 detects four or five. Since missing a battery is the more costly error in this application, we retain EfficientNet-B0 despite its higher false-alarm count. With only one EfficientNet-B3 run and three runs for each other model, we cannot tell whether differences of one or two batteries are meaningful. MobileNetV2 remains a reasonable option where false alarms matter more.

TABLE S9. COARSE-CLASSIFIER REPEATS ON THE SAME SPLITS. EACH MODEL'S THRESHOLD τ WAS SELECTED ON VALIDATION DATA TO KEEP FALSE ALARMS AT OR BELOW 3.05%. THE TABLE REPORTS TEST ACCURACY, INTACT BATTERIES MISSED (OF 259), DAMAGED BATTERIES DETECTED (OF 7), AND FALSE ALARMS (OF 426); LATENCY IS REPORTED IN THE TEXT

Model (seed)	τ	Acc.	Miss	Dmg	FA
EfficientNet-B0 (1)	0.1309	91.04%	0	4/7	19
EfficientNet-B0 (2)	0.0996	90.17%	0	5/7	21
EfficientNet-B0 (42)	0.1113	91.18%	0	5/7	21
EfficientNet-B3 (42)	0.1062	93.06%	1	6/7	16
MobileNetV2 (1)	0.1363	92.20%	1	5/7	13
MobileNetV2 (2)	0.1378	91.18%	2	4/7	12
MobileNetV2 (42)	0.1176	91.18%	1	5/7	16

The coarse classifier receives one clean image per object, while a real belt may show several items in the same frame [5], [6]. We therefore trained a YOLOv8n detector [7] on RecyBat24 bounding boxes [8] using the dataset's source-grouped split for 30 epochs. We tested it on synthetic cluttered frames: a grey belt with 3–8 non-battery items from the coarse test set and one battery crop from held-out RecyBat24 photos. We varied battery occlusion from 0% to 75% and also generated battery-free frames, with 300 frames per condition. A detector trained only on battery images marked almost every object as a battery, reporting one in 99.33% of battery-free cluttered frames. We then added 1,500 battery-free photos and 2,000 cluttered training frames made only from training-split photos. We also divided the RecyBat24 validation source photos into two groups: one generated detector validation frames, and the other supplied the test images. Thus no test battery entered training or model selection. On validation frames, the final detector achieved mAP50 = 0.995 (precision 0.993, recall 0.990). In the synthetic clutter test, it detected the battery in 99.67% of frames with no occlusion and 99.67% with 75% occlusion. The whole-frame classifier detected 7.33% and 3.33%, respectively; it was trained on single-object images and cannot isolate a small battery among other items. On battery-free frames, the detector produced a box in 1.00% and the classifier raised an alarm in 6.67% (Table S10). We count a frame as a detection when a box with confidence ≥ 0.25 has its centre inside the true battery box. The training and test frames use the same composite generator, but use different source photos. These results show performance on this kind of synthetic clutter, not expected accuracy on a real conveyor. Section S11 tests the detector on real plant frames, although the battery images are still pasted in. The whole-frame classifier is not designed for these multi-item frames. A detector front end could provide P_battery for each item to the same risk model.

TABLE S10. BATTERY DETECTION IN SYNTHETIC CLUTTER: YOLOV8N VERSUS THE WHOLE-FRAME CLASSIFIER (300 FRAMES PER CONDITION)

Condition	Detector (YOLOv8n)	Whole-frame classifier
Battery, 0% occluded: recall	99.67%	7.33%
Battery, 25% occluded: recall	100.00%	8.00%
Battery, 50% occluded: recall	99.67%	6.33%
Battery, 75% occluded: recall	99.67%	3.33%
Battery-free: frames with false alarm	1.00%	6.67%

S10. GENERALISATION AND FAIRNESS CHECKS

The original thermal model begins its phases 40–90°C below published values (Section S7). We built a second model with literature-based landmarks to check whether this choice changes the conclusions: self-heating from 90°C, separator failure from 130°C, runaway from 200°C [2], and hydrogen release from about 115°C, just below the 121°C case temperature reported at first venting [9]. We ran every system with the settings from the original tuning unchanged. Each system stopped before runaway in all 28 of 30 seeds that reached a hazard, so this model does not distinguish the systems by miss rate. The Proposed system stopped an average of 174 s before runaway, compared with 138 s for B2 and 134 s for B2-D. The same pattern held with a 20 s temperature lag. As in the original model, CUSUM stopped earliest (270 s) but missed the most hazardous time (Table IV).

We also replayed measured thermal-runaway temperature curves from the open dataset of Kwak et al. [10]. The seven cells were four cylindrical 2170 and three pouch NMC cells, both fresh and aged, heated at 5.5°C/min until runaway. This was a hybrid replay rather than a test on physical cells: measured surface temperature drove the pipeline, hydrogen and CO2 came from the calibrated gas model because the dataset has no gas channel, and camera probability came from the trained model's output on a held-out RecyBat24 test image. We ran each curve with three sensor-noise seeds. Every system stopped before the cell reached 200°C in all 21 runs. The Proposed system issued a Warning on arrival using the camera and stopped an average of 20 min before runaway; B2 stopped 18 min before. Because slow heater-driven failures

leave every method time to act, this test checks behaviour on measured curves rather than ranking the systems.

TABLE S11. MEAN STOP MARGIN BEFORE RUNAWAY WITH SETTINGS HELD FIXED: LITERATURE-CALIBRATED THERMAL MODEL (30 HELD-OUT SEEDS) AND MEASURED CELL-TEMPERATURE CURVES (7 CELLS, 3 NOISE SEEDS EACH)

System	Calibrated model: stop margin (s)	Calibrated + 20 s lag (s)	Measured-cell curves, replayed: stop margin (min)
Proposed	174	154	19.8
B2: Rule-based	138	118	17.8
B2-D: Rule + 3 s debounce	134	114	17.3
B3: Fixed Mamdani FIS	172	152	19.6
CUSUM (h = 16)	270	250	24.3

We also asked whether a model trained on the simulator could replace the hand-built rules. A logistic regression used per-step sensor features: temperature, its 5 s slope, H2, CO2, the 5 s maximum H2, and metal presence. It was trained on the tuning seeds and drove the same FSM. On the conditions represented in training, it missed no single-battery hazards and made no false escalations in the mild or hard tests. Its performance transferred poorly, however. With temperature delayed by 20 s and 60 s, it missed 0.050 and 0.254 of hazardous time, compared with 0.029 and 0.056 for the Proposed system (paired t-test p = 0.246 and 7.2e-05; 95% CIs for the difference: [−0.058, +0.014] and [−0.284, −0.122]). It also produced a false escalation for a 60°C benign item in all 30 runs, including 30 Emergency Stops, and for a warm item during gas bursts in all 30 runs. The Proposed system produced 0 and 30 escalations in those two tests, respectively. In this simulator, the physics-based rule base degrades more gradually than a model fitted to the training scenarios.

Low-cost sensors can freeze, drift, or stop reporting. We injected these failures into the held-out single-battery runs and added a health monitor. It checks for a stationary IR reading, no detected hydrogen while the cell is above 140°C, missing readings (the last value is held), a lost camera image, and a drifting gas baseline. The baseline is subtracted while the belt is empty, with a request for recalibration. When the monitor detects a fault, the controller enters Sensor_Fault, sets speed to 50%, and alerts the operator. Implementing this check exposed a problem: Sensor_Fault previously ignored risk until the fault cleared. A sensor could therefore fail during a real hazard and leave the shredder running at half speed. The controller now retains Critical or Emergency Stop if it has already reached either state, and can still escalate from Sensor_Fault. If the IR sensor freezes after arrival, the controller without the monitor misses 0.064 of hazardous time and stops only 0.6 s before runaway. With the monitor, the fault is detected 10 s after arrival; the plant runs slowly through the hazard and stops 1.9 s before runaway. The monitor flags a lost camera at arrival and detects a drifting gas sensor in 29 of 30 runs. Healthy runs produce no fault. We added 0.2°C of IR noise to every run because the simulator's empty-belt temperature has no noise. Drift compensation is not safe on a continuously full belt: learning a baseline while items pass could subtract a real gas rise when temperature is delayed.

S11. ROBUSTNESS: HAND-SET VALUES, GAS, THERMAL MODEL, REAL IMAGES

We selected the fuzzy sets, controller thresholds, dwell times, gate constants, and fusion weight α by hand. To see how sensitive the results are, we multiplied each setting by 0.8, 0.9, 1.1, and 1.2 in turn, then reran the held-out seeds using the recommended gate (src/pub_sensitivity.py; Table S12). At the paper's settings, the single-battery and 20 s-lag miss rates are 0.0064 and 0.0285. Of 120 hazardous runs, 80 stop before runaway; in this model, a cell arriving at 90°C counts as already in runaway. This is a model definition, not a general property of cells. None of the 120 battery-free stress runs falsely escalates. Five of the 12 settings produce no change within ±20%: Warning dwell, Critical dwell, corroboration temperature, gas persistence, and α. Changing the temperature sets raises the miss rate by at most 0.023, while changing the H2 sets loses one stop. The other settings have larger effects. Raising the Normal-to-Warning threshold by 20% increases the 20 s-lag miss rate to 0.1637. Lowering the Warning-to-Critical threshold by 10% or 20% causes 30 or 61 false escalations; raising C_GATE by 20% causes 37; narrowing the CO2 sets by 20% causes 15; and raising the Emergency threshold by 20% loses 6 pre-runaway stops. These values sit between missed hazards and false stops. They should be treated as commissioning settings and checked against each plant's gas-burst and temperature records, not as universal constants.

TABLE S12. ONE-AT-A-TIME SENSITIVITY OF THE RECOMMENDED GATE ON HELD-OUT SEEDS 100–129. EACH HAND-SET VALUE WAS MULTIPLIED BY 0.8, 0.9, 1.1, AND 1.2. MISS RATE IS THE LARGEST INCREASE IN SINGLE-BATTERY OR 20 S-LAG TESTS; STOPS LOST ARE HAZARDOUS RUNS (OF 120) NO LONGER STOPPED BEFORE RUNAWAY; FALSE COUNTS ARE BATTERY-FREE RUNS (OF 120) WITH A FALSE CRITICAL OR EMERGENCY STOP

Setting	Miss rate rise	Stops lost	False added
Temperature membership functions	+0.0226	+0	+0
H2 membership functions	+0.0000	+1	+0
CO2 membership functions	+0.0000	+0	+15
Normal→Warning threshold (0.30)	+0.1352	+0	+0
Warning→Critical threshold (0.55)	+0.0000	+1	+61
Critical→Emergency threshold (0.75)	+0.0000	+6	+0
Warning dwell (2 s)	+0.0000	+0	+0
Critical dwell (1 s)	+0.0000	+0	+0
Gate level C_GATE (0.5)	+0.0000	+0	+37
Corroboration temperature T_CORR (40°C)	+0.0000	+0	+0
Gas persistence (8 s)	+0.0000	+0	+0
CNN weight α (0.35)	+0.0000	+0	+0

The measured-cell datasets do not include vent-gas traces, so our gas input comes from a model. Its composition differs from published analyses. At runaway onset, the model's ratio of H2 to CO2 excess (the MQ-135 reading minus the 400 ppm ambient level, both in ppm by volume) is about 5.3. The mean across forty published vent-gas analyses is 14.4% H2 and 51.6% CO2 by volume, a ratio of 0.28 [11]; this is an average, not a range. We changed the vent-gas input in every hazardous held-out run (src/pub_gas_sensitivity.py): H2 or CO2 excess was scaled by 0.25, 0.5, or 2; the full signal was delayed by 30 or 60 s; or it was removed. These changes did not affect the single-battery or 20 s-lag miss rates because detection is carried by temperature. Scaling H2 down to match the published ratio raised the miss rates to 0.0421 and 0.0643, because with less H2 a CO2 rise remains uncorroborated for longer; every cell that had stopped before runaway still

stopped. A cell arriving at 90°C stopped 4.2 s after arrival with modelled gas and 4.1 s with no gas (30 of 30 runs), showing that temperature alone can reach Emergency Stop. This test also exposed a gap in the fuzzy temperature sets: readings above 500°C fell outside all sets, so without gas they did not trigger Emergency Stop. We extended the sets beyond that range and reran every simulation in the paper. The main tables changed by no more than 0.1 s.

Section S7 showed that the main thermal model begins its phases 40–90°C below published values. We ran the same seeds using the literature-calibrated model (src/pub_thermal_compare.py). The main model is the harder case: its mean stop margin is 22, 17, and 8 s with no IR lag and with 20 and 60 s of lag, respectively. The corresponding margins are 173, 153, and 113 s for the calibrated model, where every run stops before runaway. Thus, the main-model results are more conservative. With a 60 s IR lag on the main model, B2 stops before runaway in 27 of 27 runs and the Proposed system in 23 of 27. That difference reflects the Proposed system's refusal to stop on every gas burst.

We tested the camera on three external image datasets (Table S13). BatSort and XBAT+ were not used for training or model selection. For ZeroWaste, we later used frames from 9 videos to fine-tune the detector and held out the other 7 videos for evaluation. BatSort [5] is a Singapore battery dataset containing web images of ten battery types, including Ni-MH and Ni-Cd cells absent from our training data. Before testing, we removed images whose perceptual hashes were too close to any of our own photos. At the divert threshold, the coarse model labels 176 of the remaining 179 images as Battery, including all Ni-MH and Ni-Cd photos (8 of 8 and 7 of 7; with near-duplicates retained, 21 of 21 and 24 of 24). Because diversion uses the coarse Battery class, it is not limited to Li-ion cells.

XBAT+ [12] contains webcam images of battery-containing devices sorted from Irish WEEE collection drums. We used all 421 RGB images in its training and test splits; none was used to train our models. We excluded one near-duplicate of our own photos, leaving 420 images. The camera alone would divert 348 of them (83%), as reported in Table S13. Phones and remote controls make up most of the set (246 of 420), and their battery or compartment is often visible. In comparison, only 3.7% of the Cardboard_Trash images used to represent hidden batteries in Section V-E cross the threshold.

ZeroWaste [6] contains images from a working recycling conveyor, labelled as cardboard, soft plastic, rigid plastic, or metal; it has no battery class. On 136 unmodified frames, the YOLOv8n detector from Section S9 marked a battery in 17.6%, compared with 1.0% on composite battery-free frames. It mistook flat white paper and card for pouch cells. We fine-tuned the detector for 10 epochs on 264 frames from 9 ZeroWaste videos, each used once without a battery and once with a pasted RecyBat24 battery from the training split. We evaluated both the original and fine-tuned detectors on frames from the other 7 videos because frames from one video are nearly identical. On the held-out real frames, fine-tuning reduced false boxes to 0.7%. It detected pasted held-out batteries, taken from the unused half of RecyBat24 validation photos as in Section S9, in 99.3% of uncovered frames and 100.0% of frames with 75% occlusion. Before fine-tuning, the rates were 99.3% and 97.1%. The original validation frames gave mAP50 = 0.995. These are genuine conveyor images, but the battery images pasted into them are not real batteries on the belt.

We also tested 44 freely licensed Wikimedia Commons photos selected to represent items a sorting line might encounter. The camera detected 22 of 26 batteries, flagged all 6 battery-containing devices, and correctly ignored 4 of 12 other objects. It missed 2 of 5 swollen or damaged batteries and flagged some cell-shaped objects, including electrolytic capacitors (results/external/demo_photos_summary.txt).

TABLE S13. CAMERA AND DETECTOR PERFORMANCE ON EXTERNAL IMAGES. BATSORT AND XBAT+ WERE NOT USED FOR TRAINING; ZEROWASTE EVALUATION USED 7 VIDEOS HELD OUT FROM FINE-TUNING. BATTERY THRESHOLD: P_BATTERY ≥ 0.1113; YOLOV8N BOX CONFIDENCE ≥ 0.25. BATSORT PER-TYPE RESULTS INCLUDE NEAR-DUPLICATES

Images (n)	Result
BatSort [5], Ni-MH (21)	21 detected as Battery
BatSort [5], Ni-Cd (24)	24 detected as Battery
BatSort [5], Li-ion 18650 (51)	51 detected as Battery
BatSort [5], lithium primary (33)	33 detected as Battery
BatSort [5], alkaline (153)	150 detected as Battery
BatSort [5], alkaline (industrial) (19)	19 detected as Battery
BatSort [5], other (59)	58 detected as Battery
BatSort, all types, near-duplicates removed (179)	176 detected (98.3%)
XBAT+ [12], real devices holding a battery (420)	348 diverted by the camera alone
ZeroWaste [6] real frames, no battery, detector v2 (136)	17.6% frames with a false box
ZeroWaste + pasted battery, 0 / 75% covered, detector v2 (136)	99.3% / 97.1% found
ZeroWaste [6] real frames, no battery, fine-tuned on other videos (136)	0.7% frames with a false box
ZeroWaste + pasted battery, 0 / 75% covered, fine-tuned on other videos (136)	99.3% / 100.0% found

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